

Consumer theory: duality

In the previous chapter we defined the consumer problem as that of choosing a vector x to solve the problem $\max u(x)$ s.t. $px = M$, where p is a price vector and M money income. From the solution we derived *Marshallian* demand functions $x_i = D_i(p, M)$ ($i = 1, \dots, n$), which express demands as functions of prices and money income. We observed that we cannot place restrictions on the signs of the partial derivatives of these functions: $\partial D_i / \partial M \geq 0$, $\partial D_i / \partial p_j \leq 0$ ($i, j = 1, \dots, n$). In particular the demand for a good does not necessarily vary inversely with its own price. However, as a result of a diagrammatic analysis, we *were* able to say that this will be true of normal goods, or of inferior goods whose income effects are weaker than their substitution effects. We now put this analysis on a more rigorous and general basis. We also consider the problem, central to many applications of consumer theory, of deriving a money measure of the costs and benefits incurred by a consumer as a result of price changes. In doing so, we develop the methods and concepts of *duality theory*, an approach to the analysis of optimization problems which permits an elegant and concise derivation of comparative static results.

A. The expenditure function

The expenditure function is derived from the problem of minimizing the total expenditure necessary for the consumer to achieve a specified level of utility u :

$$\begin{aligned} \min_{x_1, \dots, x_n} \sum p_i x_i \quad \text{s.t.} \quad & \text{(i) } u(x_1, \dots, x_n) \geq u \\ & \text{(ii) } x_i \geq 0, \quad i = 1, \dots, n \end{aligned} \quad [\text{A.1}]$$

If all prices are positive the first constraint in [A.1] will be satisfied as an equality in the solution, since if $u(x) > u$ expenditure can be reduced without violating the constraint. If it is further assumed that all x_i are strictly positive in the solution, we can write the Lagrange function for the problem (with μ as the Lagrange multiplier) as

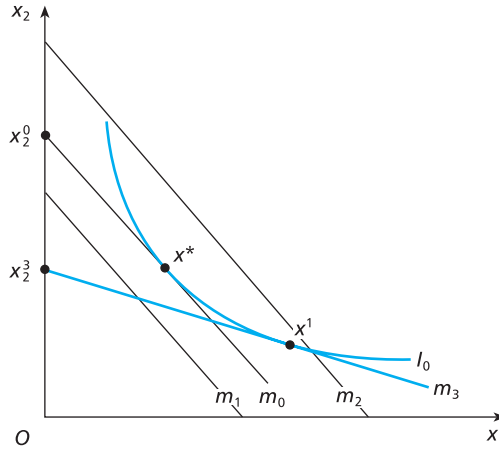
$$L = \sum p_i x_i + \mu[u - u(x_1, \dots, x_n)] \quad [\text{A.2}]$$

and the necessary conditions for a minimum of L , also the necessary conditions for a solution of [A.1], are

$$\frac{\partial L}{\partial x_i} = p_i - \mu u_i = 0 \quad i = 1, \dots, n \quad [\text{A.3}]$$

$$\frac{\partial L}{\partial \mu} = u - u(x_1, \dots, x_n) = 0 \quad [\text{A.4}]$$

Figure 3.1



The conditions on the x_i bear a striking resemblance to [C.9] in Chapter 2. Writing them as $p_i = \mu u_i$ and dividing the condition on x_i by the condition on x_j gives

$$\frac{p_i}{p_j} = \frac{u_i}{u_j} \quad [\text{A.5}]$$

which is identical with Chapter 2 [C.2]: the ratio of prices is equated to the marginal rate of substitution. This is not surprising as examination of the two-good case in Fig. 3.1 indicates. The indifference curve I_0 shows the combinations of x_1 and x_2 which give a utility level of u and the feasible set for the problem is all points on or above I_0 . The lines m_0, m_1, m_2 , are isoexpenditure lines similar to the budget lines of earlier diagrams. m_0 , for example, plots all bundles costing m_0 , i.e. satisfying the equation $p_1 x_1 + p_2 x_2 = m_0$. The problem is to find the point in the feasible set which is on the lowest isoexpenditure line. This will, in the tangency solution shown here, be where the indifference curve I_0 is tangent to the isoexpenditure line m_0 . The problem confronting the *utility-maximizing* consumer is to move along the budget line until the highest indifference curve is reached. The *expenditure-minimizing* problem is to move along the indifference curve until the lowest isoexpenditure line is reached.

The optimal x_i^* in problem [A.1] depend on the prices and the utility level u :

$$x_i^* = H_i(p_1, \dots, p_n, u) = H_i(p, u) \quad i = 1, \dots, n \quad [\text{A.6}]$$

and $H_i(p, u)$ is the *Hicksian* demand function for x_i . Substituting the optimal values of the x_i in $\sum p_i x_i$ gives

$$\sum p_i x_i^* = \sum p_i H_i(p, u) = m(p, u) \quad [\text{A.7}]$$

$m(p, u)$ is the *expenditure function*, showing the minimum level of expenditure necessary to achieve a given utility level as a function of prices and the required utility level.

The Hicksian demand function is also called the *compensated demand function*. In considering the effect of a change in price on demand with utility held constant (the partial derivative $\partial H_i / \partial p_j$ ($i, j = 1, \dots, n$)) we automatically make whatever changes in expenditure are required to compensate for the effects of the price

change on real income or utility. This is illustrated in Fig. 3.1. Assume p_2 remains constant while p_1 falls to give a new family of isoexpenditure lines, with slopes corresponding to that of m_3 in the figure. x^1 is the new expenditure minimizing consumption bundle, and the change from x^* to x^1 is the effect of making the relative price change with m varying to keep u constant. The optimal expenditure line slides round the indifference curve from m_0 to m_3 as the optimal bundle changes from x^* to x^1 . The minimized total expenditure can be read off from the intercepts of m_0 and m_3 on the x_2 axis. The fall in p_1 lowers m from $p_2x_2^0$ to $p_2x_2^3$.

Provided the indifference curves are strictly convex to the origin the optimal x_i (and hence the expenditure function) vary smoothly and continuously with the prices of the goods. Hence the $H_i(p, u)$ functions have continuous derivatives with respect to the prices. The demand curve we derive from the Hicksian demand function was represented by curve hh in Fig. 2.15. The slope of the Hicksian or compensated demand curve, $\partial H_i / \partial p_i$ ($i = 1, \dots, n$), is the *substitution effect of the price change*, since by definition $\partial H_i / \partial p_i$ is taken with u held constant.

The expenditure function gives the smallest expenditure, at a given price vector, that is required to achieve a particular ‘standard of living’ or utility level, and describes how that expenditure will change as prices or the required utility level change. The assumptions made in Chapter 2 concerning the nature of the consumer’s preference ordering and indifference sets imply certain properties of the expenditure function:

(a) The expenditure function is concave in prices

Choose two price vectors p' and p'' , and k such that $0 \leq k \leq 1$. Define $\bar{p} = kp' + (1 - k)p''$. We have to prove that (see the definition of concavity in Appendix B):

$$m(\bar{p}, u) \geq km(p', u) + (1 - k)m(p'', u)$$

for given u .

Proof

Let x' and x'' solve the expenditure minimization problem when the price vector is respectively p' and p'' . By definition of the expenditure function, $p'x' = m(p', u)$ and $p''x'' = m(p'', u)$. Likewise, let $\bar{x}$ solve the problem when the price vector is $\bar{p}$, so that $\bar{p}\bar{x} = m(\bar{p}, u)$. Since x' and x'' are solutions to their respective expenditure minimization problems we must have

$$p'\bar{x} \geq p'x' \quad \text{and} \quad p''\bar{x} \geq p''x'' \quad [\text{A.8}]$$

Multiplying through the first inequality by k and the second by $1 - k$ and summing, gives

$$kp'\bar{x} + (1 - k)p''\bar{x} \geq kp'x' + (1 - k)p''x'' \quad [\text{A.9}]$$

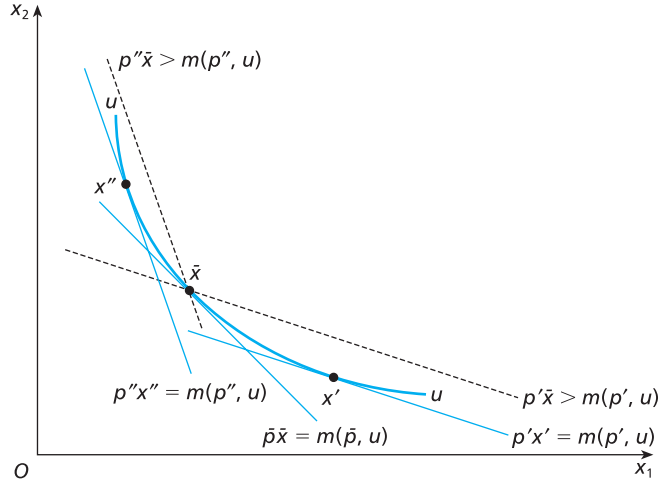
But by definition of $\bar{p}$ this implies

$$(kp' + (1 - k)p'')\bar{x} = \bar{p}\bar{x} \geq kp'x' + (1 - k)p''x'' \quad [\text{A.10}]$$

which is the result we want.

Figure 3.2 illustrates the proof of this important result. It is obvious that, when the isoexpenditure lines at which x' and x'' are optimal solutions are shifted so as to pass through point $\bar{x}$, they must yield higher expenditure, thus giving the key inequalities in [A.8]. The rest of the proof then follows by simple algebra.

Figure 3.2



The figure could in one sense be misleading. The inequalities (which in this case are *strict*) appear to follow from the convexity of the indifference curves. Note, however, that the inequalities in [A.8] follow simply from the fact that x' (respectively x'') minimizes px at price vector p' (respectively p'') while $\bar{x}$ may not – [A.8] then follows from the definition of a minimum. Thus the proof of concavity of the expenditure function does *not depend* on convexity of preferences. However, the property of *uniqueness* of solutions like x' and x'' , and the differentiability of Hicksian demands and of the expenditure function, do. Note that strict convexity of preferences implies strict concavity of the expenditure function at an interior solution to problem [A.1], since it implies uniqueness of the solution and hence strict inequalities in [A.8].

Figure 3.3 illustrates the strict concavity of the expenditure function when the price vectors p' and p'' differ only in respect of one price, p_i . The slope of the expenditure function at a point is equal to the compensated demand for good i at the price p_i :

(b) Shephard's lemma: $\partial m(p, u)/\partial p_i = x_i^* = H_i(p, u)$

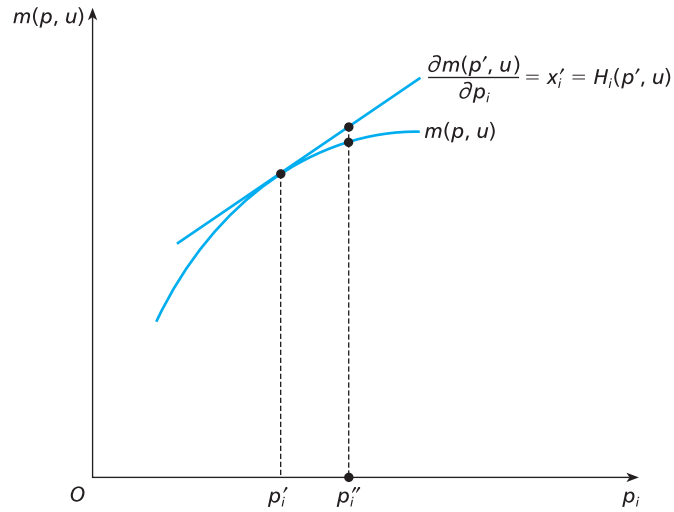
The proof is just a version of the Envelope Theorem (Appendix J). Differentiating [A.7] with respect to the i th price gives

$$\frac{\partial m}{\partial p_i} = x_i^* + \sum_{j=1}^n p_j \frac{\partial x_j^*}{\partial p_i} = x_i^* + \mu \sum_{j=1}^n u_j \frac{\partial x_j^*}{\partial p_i} = x_i^* \quad [\text{A.11}]$$

The second equality uses the fact that $p_i = \mu u_i$ from the first-order condition [A.3]. Since utility is held constant when p_i varies, differentiating the constraint [A.4] with respect to p_i shows that $\sum_{j=1}^n u_j \partial x_j^* / \partial p_i = 0$ which gives the third equality in [A.11].

Thus the partial derivative of the expenditure function with respect to the i th price is the compensated demand for the i th good. In Fig. 3.3, the slope of the curve at price p'_i is $x'_i = H_i(p'_1, \dots, p'_i, \dots, p'_n, u)$. This can be put intuitively as follows. Suppose a consumer buys 12.5 units of gas a week at a cost of £1 per unit. The price

Figure 3.3



of gas then rises by 1p per unit. Shephard's lemma says that, *to a first approximation*, to maintain the same utility level or standard of living her expenditure must increase by $H_i \Delta p_i = 12.5p$: just enough to maintain consumption at the initial price level. The qualification 'to a first approximation' is important. For *finite* price changes Fig. 3.3 shows that $H_i \Delta p_i$ overstates the required increase in expenditure, since the expenditure function is strictly concave. As a good's price goes up, the consumer substitutes *away* from the good in question, and this reduces the amount of expenditure otherwise required to keep utility constant. Recall from Chapter 2 the distinction between Hicks and Slutsky compensated demands. Shephard's lemma tells us that for small enough price changes this distinction can be ignored.

(c) $\partial m / \partial p_i \geq 0$ with strict inequality if $x_i^* > 0$

This follows immediately from Shephard's lemma. Since at least one good must be bought, the expenditure function is non-decreasing in the price vector p and strictly increasing in at least one price. Higher prices mean higher expenditure to reach a given utility.

(d) The expenditure function is homogeneous of degree 1 in prices

Take a given u value and price vector p^0 , and let $m^0 = m(p^0, u) = p^0 x^0$ where x^0 is the expenditure-minimizing bundle at p^0 , that is $p^0 x^0 \leq p^0 x$ for all bundles x yielding utility of u or more. But this implies that $k p^0 x^0 \leq k p^0 x$ for all bundles yielding at least u and so x^0 is optimal at prices p^0 and $k p^0$. Then $m(k p^0, u) = k p^0 x^0 = k m(p^0, u)$. Since *relative* prices do not change, the optimal bundle is not changed: it has merely become more or less expensive depending on whether $k > 1$ or $k < 1$.

(e) The expenditure function is increasing in u

Higher utility at given prices requires higher expenditure. Rather than use the envelope theorem again, recall that the Lagrange multiplier $\mu > 0$ in [A.2] is equal to the

derivative $\partial m/\partial u$ (see Appendix G). μ is the ‘marginal cost of utility’, since it represents the rate of change of *minimized* expenditure with respect to the required utility level. μ is the reciprocal of the Lagrange multiplier λ in the corresponding utility maximization problem, i.e. μ is the inverse of the ‘marginal utility of income’ (see Question 3, Exercise 3A). Note that, although the assumptions underlying ordinal utility theory allow the sign of μ to be established, we cannot say that μ is necessarily increasing, or decreasing, with u , because both are possible for different, permissible utility functions (see Question 3, Exercise 3A).

It is important to be clear about the relation between expenditure and utility. The essential facts about the consumer’s preference ordering are contained in the structure of her indifference sets or curves. The minimum expenditure required to reach a given indifference set at given prices is unaffected by any number we attach to that indifference set to indicate its place in the ordering. On the other hand, once we have chosen a numerical representation of the preference ordering – a utility function – this will imply a particular relationship between expenditure m and utility u . But the properties we set out above hold for all permissible utility functions, and the only general restriction we can place on the relation between m and u (for a given price vector) is that it is monotonically increasing.

EXERCISE 3A

1. *Cobb–Douglas utility function.* A consumer has the utility function $u = x_1^a x_2^b$, $a + b = 1$. Derive her Hicksian demand functions and expenditure function. Confirm that the expenditure function possesses the properties set out in this section. Then derive the expenditure function for the utility function $v = u^2$ and compare it with the one you obtained previously. In particular, compare the values $\partial m/\partial u$ in each case.
2. If goods are perfect complements the consumer’s utility function can be written: $u = \min(x_1, x_2)$. If the goods are perfect substitutes the utility function can be written $u = ax_1 + bx_2$. Discuss the nature of the expenditure function in each of these cases.
3. Consider the problems:

$$\max u(x) \quad \text{s.t. } px = m; \quad \min px \quad \text{s.t. } u = u(x)$$

where u is a strictly quasi-concave utility function and p is the same price vector in each case. With m given, let u^* be the optimized utility in the first problem, with every $x_i^* > 0$. Then let u^* be the value of the utility constraint in the second problem. Then show:

- (a) the solution vector in the second problem is identical to that in the first;
 - (b) $\lambda^* = 1/\mu^*$, where λ^* and μ^* are the optimal values of the Lagrange multipliers in the first and second problems respectively;
 - (c) these results hold for any positive monotonic transformation of the utility function.
4. *Stone–Geary utility function.* A consumer has the utility function

$$u = (x_1 - c_1)^a (x_2 - c_2)^b, \quad a + b = 1$$

where the c_i are interpreted as minimum subsistence levels of x_i , $i = 1, 2$. Derive the consumer’s Hicksian demand functions and expenditure function, and compare them with the results obtained in Question 1.

5. *Quasi-linear utility function.* A consumer has the utility function

$$u = f(x_1) + x_2$$

where $f(\cdot)$ is increasing and strictly concave. Derive the Hicksian demand functions and expenditure function and discuss their properties. Illustrate on an indifference curve diagram.

B. The indirect utility function, Roy's identity and the Slutsky equation

The indirect utility function is derived from the consumer problem of maximizing $u(x_1, \dots, x_n)$ subject to the budget constraint $\sum p_i x_i \leq M$ and non-negativity constraints. We saw in section 2D that the x_i which are optimal for this problem will be functions of the p_i and M : $x_i^* = D_i(p_1, \dots, p_n, M) = D_i(p, M)$. The *maximized* value of $u(x_1, \dots, x_n) = u(x_1^*, \dots, x_n^*)$ will therefore also be a function of the p_i and M :

$$\begin{aligned} u(x_1^*, \dots, x_n^*) &= u(D_1(p, M), \dots, D_n(p, M)) \\ &= u^*(p, M) \end{aligned} \quad [\text{B.1}]$$

u^* is known as the *indirect utility function* since utility depends indirectly on prices and money income via the maximization process, in contrast to the utility function $u(x_1, \dots, x_n)$ where u depends directly on the x_i . We can use u^* to investigate the effects of changes in prices and money income on the consumer's utility.

From the interpretation of the Lagrange multiplier (section 2B), the effect of an increase in money income on the maximized utility is

$$\frac{\partial u^*}{\partial M} = \lambda \quad [\text{B.2}]$$

The effect of a change in p_i on u^* can also be found as a version of the Envelope Theorem. Differentiating u^* with respect to p_i :

$$\frac{\partial u^*}{\partial p_i} = \sum u_k \frac{\partial x_k^*}{\partial p_i} = \lambda \sum p_k \frac{\partial x_k^*}{\partial p_i} \quad [\text{B.3}]$$

The budget constraint must still be satisfied so that

$$\frac{d}{dp_i} \left(\sum p_k x_k^* \right) = \frac{dM}{dp_i} = 0$$

and so

$$\sum p_k \frac{\partial x_k^*}{\partial p_i} + x_i^* = 0$$

or

$$-x_i^* = \sum p_k \frac{\partial x_k^*}{\partial p_i}$$

Substitution of this in [B.3] gives Roy's identity:

$$\frac{\partial u^*}{\partial p_i} = -\lambda x_i^* = -\frac{\partial u^*}{\partial M} x_i^* \quad [\text{B.4}]$$

The expression on the right-hand side of [B.4] has the following intuitive explanation. An increase in p_i is a reduction in the purchasing power of the consumer's money income M , and by Shephard's lemma, to the first order, her purchasing power falls at the rate $-x_i^*$ as p_i varies. λ is the marginal utility of money income. The product of λ and $-x_i^*$ is the rate at which utility varies with money income, times the rate at which (the purchasing power of) money income varies with p_i , and so this product yields the rate of change of utility with respect to p_i .

Since $\lambda > 0$, Roy's identity shows that an increase in the price of a good a consumer buys reduces her (maximized) utility or standard of living to a greater extent, the larger the quantity of it she buys.

The indirect utility function tells us that utility depends, via the maximization process, on the price-income situation the consumer faces. Note that [B.2] implies that the indirect utility function is monotonically increasing in income, M . Thus we can invert the indirect utility function $u = u^*(p, M)$ to obtain the expenditure function $M = m(p, u)$. A given solution point for a given price vector can be viewed equivalently as resulting from minimizing expenditure subject to the given utility level or maximizing utility subject to the given expenditure level. We can choose either to solve the utility maximization problem, obtain the indirect utility function and invert it to obtain the expenditure function, *or* to obtain the expenditure function and then invert it to obtain the indirect utility function (see Question 3, Exercise 3B). The two functions are *dual to each other*, and contain essentially the same information: the forms of the functions and their parameters are completely determined by the form of the original (direct) utility function. But then, since each of these three functions contains the same information, we can choose any one of them as the representation of the consumer's preferences that we wish to work with.

Duality can be used to give a neater derivation of Roy's identity. Setting $M = m(p, u)$, rewrite the indirect utility function as

$$u = u^*(p, m(p, u)) \quad [\text{B.5}]$$

Then differentiating through with respect to p_i , allowing m to vary in such a way as to hold u constant, gives

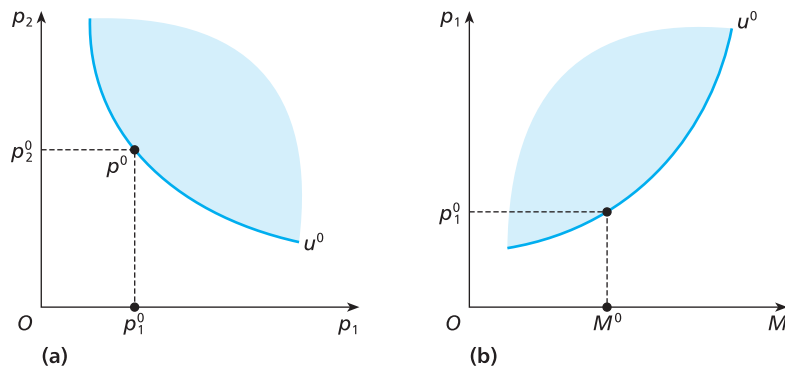
$$0 = \frac{\partial u^*}{\partial p_i} + \frac{\partial u^*}{\partial M} \frac{\partial m}{\partial p_i} \quad [\text{B.6}]$$

which, using Shephard's lemma and [B.2], gives Roy's identity [B.4] directly.

Since the indirect utility function is ordinal and not cardinal, we cannot restrict it to be convex or concave (unlike the expenditure function), but we can show that it is *quasi-convex* in prices and income, a property that is useful in many applications.

Figure 3.4 illustrates quasi-convexity in prices and income. A function is quasi-convex if, given any point in its (convex) domain, the *worse set* of that point, i.e. the

Figure 3.4



set of points giving the same or lower values of the function, is convex. Take the case of two goods, where the indirect utility function is $u^*(p_1, p_2, M)$. In part (a) of the figure, $p^0 = (p_1^0, p_2^0)$ is some arbitrary point, and the indifference curve u^0 , or contour of the indirect utility function, through that point is convex to the origin if the function is quasi-convex. The worse set $W(p^0) = \{(p_1, p_2); u^*(p_1, p_2, M^0) \leq u^*(p_1^0, p_2^0, M^0)\}$ lies to the north-east of p^0 (higher prices imply lower utility) and is convex. In (b) of the figure, the contour u^0 through the point (p_1^0, M^0) is convex from below, because the worse set $W(p_1^0, M^0) = \{(p_1, M); u^*(p_1, p_2^0, M) \leq u^*(p_1^0, p_2^0, M^0)\}$ lies to the north-west of the point (higher price and lower income implies lower utility) and is convex. (Be sure you can explain the negative and positive slopes of these contours, respectively.) Similarly for any point (p_2^0, M^0) .

To prove that the indirect utility function is quasi-convex in prices and income, choose two points in the domain of the function, (p^0, M^0) and (p', M') , such that

$$u^*(p^0, M^0) = u^0 \geq u' = u^*(p', M') \quad [\text{B.7}]$$

so that (p', M') is in the worse set of (p^0, M^0) . We have to show that any convex combination of these two price-income vectors is also in this worse set of (p^0, M^0) :

$$u^*(\bar{p}, \bar{M}) \leq u^0 \quad [\text{B.8}]$$

where

$$\bar{p} = kp^0 + (1 - k)p', \quad \bar{M} = kM^0 + (1 - k)M' \quad k \in [0, 1] \quad [\text{B.9}]$$

Now take any goods vector x that satisfies the budget constraint $\bar{p}x \leq \bar{M}$ or, given the definitions of $\bar{p}$ and $\bar{M}$

$$kp^0x + (1 - k)p'x \leq kM^0 + (1 - k)M' \quad [\text{B.10}]$$

For this to hold either

$$p^0x \leq M^0 \quad [\text{B.11}]$$

or

$$p'x \leq M' \quad [\text{B.12}]$$

or both. Now x satisfying these inequalities cannot yield a higher utility value than the maximized utility at the corresponding budget constraint. Hence [B.11] implies $u^*(\bar{p}, \bar{M}) \leq u^0$, and [B.12] implies $u^*(\bar{p}, \bar{M}) \leq u'$ and since one or both of [B.11] and [B.12] must hold and $u' \leq u^0$ by assumption, we have established $u^*(\bar{p}, \bar{M}) \leq u^0$ as required.

Figure 3.5 illustrates. Fix the price of good 2 as 1, so that the intercept on the x_2 axis shows total expenditure and the slope of the budget constraint is $-p_1$. B^0 in the figure corresponds to the budget constraint $p_1^0x_1 + x_2 = M^0$. B' corresponds to $p_1'x_1 + x_2 = M'$, and yields a lower utility value than B^0 . $\bar{B}$ corresponds to $\bar{p}_1x_1 + x_2 = \bar{M}$ where $\bar{p}_1 = kp_1^0 + (1 - k)p_1'$, $\bar{M} = kM^0 + (1 - k)M'$. $\bar{B}$ also yields a lower value of utility than B^0 . We have

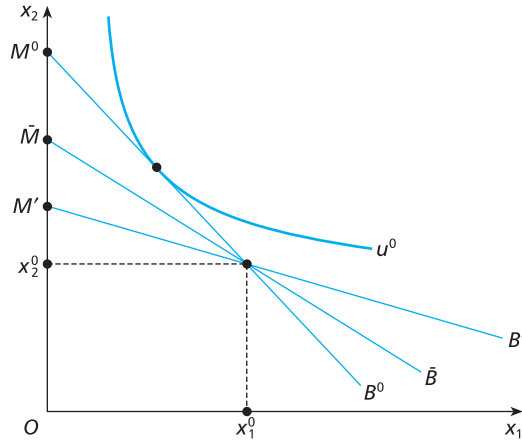
$$M' < \bar{M} < M^0 \quad [\text{B.13}]$$

$$p_1' < \bar{p}_1 < p_1^0 \quad [\text{B.14}]$$

That $\bar{B}$ passes through the intersection point (x_1^0, x_2^0) of B^0 and B' follows by noting that if we sum

$$k(p_1^0x_1^0 + x_2^0) = kM^0 \quad [\text{B.15}]$$

Figure 3.5



and

$$(1 - k)(p'_1 x_1^0 + x_2^0) = M' \quad [\text{B.16}]$$

we obtain

$$\bar{p}_1 x_1^0 + x_2^0 = \bar{M} \quad [\text{B.17}]$$

Thus (p', M') and $(\bar{p}, \bar{M})$ are both in the worse set of (p^0, M^0) and $(\bar{p}, \bar{M})$ is a convex combination of (p^0, M^0) and (p', M') .

The Slutsky equation

The Slutsky equation plays a central role in analysing the properties of demand functions. It is derived as follows. If we take as the constraint in the utility maximization problem the level of expenditure resulting from solution of the expenditure minimization problem (or equivalently take as the constraint in the latter problem the level of utility resulting from the solution to the former) then the solutions x_i^* to the two problems, the values of the Marshallian and Hicksian demand functions, will be identical. Setting $M = M(p, u)$, we can write for the i th good

$$H_i(p, u) = D_i(p, m(p, u)) \quad [\text{B.18}]$$

Since [B.18] is an identity we can differentiate through with respect to the j th price, allowing expenditure to change in whatever way is required to keep utility constant, to obtain

$$\frac{\partial H_i}{\partial p_j} = \frac{\partial D_i}{\partial p_j} + \frac{\partial D_i}{\partial M} \frac{\partial m}{\partial p_j} \quad [\text{B.19}]$$

Using Shephard's lemma and rearranging gives the *Slutsky equation*

$$\frac{\partial D_i}{\partial p_j} = \frac{\partial H_i}{\partial p_j} - x_j \frac{\partial D_i}{\partial M} \quad [\text{B.20}]$$

Taking $i = j$, so that we consider the effect of a price change on its own demand, we see from [B.20] that the slope of the Marshallian demand function is the sum of two effects: the *substitution effect*, $\partial H_i / \partial p_i$, which is the slope of the Hicksian or compensated demand curve; and the *income effect*, $-x_i \partial D_i / \partial M$. Thus the Slutsky equation gives a precise statement of the conclusions of the diagrammatic analysis of Chapter 2. We show in a moment that $\partial H_i / \partial p_i \leq 0$. Then [B.20], again with $i = j$, establishes that if the good is normal, so that $\partial D_i / \partial M > 0$, the slope of its Marshallian demand curve is negative. If the good is inferior, so that $\partial D_i / \partial M \leq 0$, the slope is negative, positive or zero depending on the relative sizes of the absolute values $|\partial H_i / \partial p_i|$ and $|x_i \partial D_i / \partial M|$.

It is useful to express the Slutsky equation in elasticity form. Again taking $i = j$, multiplying through [B.20] by p_i/x_i , and the income term by M/M , gives

$$\varepsilon_{ii} = \sigma_{ii} - s_i \eta_i \quad [\text{B.21}]$$

where ε_{ii} is the Marshallian demand elasticity, σ_{ii} is the Hicksian or compensated demand elasticity, η_i is the income elasticity of demand, and $s_i = p_i x_i / M$ is the share of good i in total expenditure. Thus the difference between Hicksian and Marshallian elasticities for a good will be smaller, the smaller its income elasticity and the less significant it is in the consumer's budget. With $i \neq j$, [B.20] becomes

$$\varepsilon_{ij} = \sigma_{ij} - s_j \eta_i \quad [\text{B.22}]$$

which emphasizes that cross-price Marshallian demand elasticities depend both on compensated elasticities and on income elasticities weighted by expenditure shares. Equality of the Marshallian cross-price elasticities therefore requires strong restrictions on preferences (see Question 5, Exercise 3B).

We define the *Slutsky matrix* as the $n \times n$ matrix $[\partial H_i / \partial p_j]$ of Hicksian demand derivatives. It is a straightforward extension of Shephard's lemma and the properties of the expenditure function to show that this matrix is a *symmetric, negative semi-definite* matrix (Appendix I). From Shephard's lemma

$$\frac{\partial m(p, u)}{\partial p_i} = H_i(p, u) \quad i = 1, \dots, n$$

we have

$$\frac{\partial^2 m(p, u)}{\partial p_j \partial p_i} = \frac{\partial H_i}{\partial p_j} \quad i, j = 1, \dots, n \quad [\text{B.23}]$$

Then, from *Young's Theorem*¹ we have immediately that $\partial H_i / \partial p_j = \partial H_j / \partial p_i$, and so the Slutsky matrix is symmetric. The Slutsky matrix $[\partial H_i / \partial p_j]$ is the matrix of second-order partials of the expenditure function and the concavity of the expenditure function implies that matrix is negative semi-definite. Since $\partial^2 m / \partial p_i^2 = \partial H_i / \partial p_i \leq 0$, by the definition of negative semi-definiteness (see Appendix I), the Hicksian demand curve cannot have a positive slope. We have seen earlier that strict convexity of preference and $x_i > 0$ at the optimum establish the stronger result that $\partial^2 m / \partial p_i^2 = \partial H_i / \partial p_i < 0$.

The Hicksian demand derivative $\partial H_i / \partial p_j$ is often used to define complements and substitutes. Two goods i and j are called *Hicksian complements* if $\partial H_i / \partial p_j < 0$ and *Hicksian substitutes* if $\partial H_i / \partial p_j > 0$. The advantage of this definition is that symmetry implies that the nature of the complementarity or substitutability between the goods cannot change if we take $\partial H_j / \partial p_i$ rather than $\partial H_i / \partial p_j$. This would *not* be true

if we defined complements and substitutes in terms of the Marshallian demand derivatives (see Question 5, Exercise 3B).

Properties of demand functions

We have seen that it is possible to draw definite conclusions about the effects of price changes on the Hicksian demands. The Hicksian demand functions are not, however, directly observable since they depend on the consumer's utility level as well as prices. On the other hand, the Marshallian demand functions can be estimated from information on purchases, prices and money income. The Slutsky equation enables us to reformulate the predictions about the properties of Hicksian demand functions in terms of the observable Marshallian demand functions and thus to widen the set of testable predictions from consumer theory.

We can summarize the testable implications derived in this and the previous chapter:

- (a) Marshallian demand functions are homogeneous of degree zero in prices and money income;
- (b) the Marshallian demand functions satisfy the 'adding up' property: $\sum p_i x_i^* = M$;
- (c) the Hicksian demand derivatives (cross-substitution effects) are symmetric: $\partial H_i / \partial p_j = \partial H_j / \partial p_i$ or, using the Slutsky equation, $\partial D_i / \partial p_j + x_j \partial D_i / \partial M = \partial D_j / \partial p_i + x_i \partial D_j / \partial M$;
- (d) the Slutsky matrix $[\partial H_i / \partial p_j] = [\partial D_i / \partial p_j + x_j \partial D_i / \partial M]$ is negative semi-definite.

These are all the predictions about the Marshallian demand functions which can be made on the basis of the consumer preference axioms. (As we will see in sections D and E, more detailed predictions require stronger and less general specifications of preferences.) The converse question of whether a system of demand functions with these properties implies the existence of a utility function from which the demand functions could have been derived is known as the *integrability problem*. In the next section we will show that this is in fact so by considering the equivalent problem of retrieving an expenditure function (which also can be used to represent preferences) from a set of Marshallian demand functions which satisfy the above properties.

EXERCISE 3B

1. Show that the Hicksian demand function is homogeneous of degree zero in prices. Then, use the fact (Euler's Theorem) that if a function $f(x_1, \dots, x_n)$ is homogeneous of degree zero, we have $\sum_{i=1}^n f_i x_i = 0$, to prove that $\sum_{j=1}^n (\partial H_i / \partial p_j) p_j = 0$. Interpret this in terms of the Slutsky matrix.
2. The consumer has the utility function $u = x_1^a x_2^{1-a}$. Find her indirect utility function. Confirm Roy's identity by:
 - (a) differentiating the indirect utility function with respect to the price of good 1;
 - (b) using the first-order conditions to obtain solutions for x_1 and λ , and therefore an expression for $-\lambda x_1$;
 - (c) showing that (a) and (b) give the same result.
3. Invert the indirect utility function you obtain in Question 2 to express expenditure as a function of prices and utility. Then show that this is the expenditure function for this form of direct utility function.

4. (a) Show that the Marshallian demand functions satisfy the following restrictions:

$$\text{Cournot aggregation: } \sum_{i=1}^n p_i \frac{\partial D_i}{\partial p_j} + x_j = 0 \quad j = 1, \dots, n$$

$$\text{Engel aggregation: } \sum_{i=1}^n p_i \frac{\partial D_i}{\partial M} = 1$$

(Hint: use the adding up property and differentiate.)

- (b) Express these restrictions in elasticity form

$$\begin{aligned} \sum_i s_i e_{ij} + s_j &= 0 \quad j = 1, \dots, n \\ \sum_i s_i \eta_i &= 1 \end{aligned}$$

where e_{ij} is the cross-price elasticity $(\partial D_i / \partial p_j)(p_j / x_i)$, η_i is the income elasticity $(\partial D_i / \partial M)(M / x_i)$ and $s_i = p_i x_i / M$ is the budget or expenditure share of the i th good.

- (c) Show that the homogeneity property implies

$$\sum_{j=1}^n p_j \frac{\partial D_i}{\partial p_j} + M \frac{\partial D_i}{\partial M} = 0 \quad i = 1, \dots, n$$

and express this in elasticity form

$$\sum_j e_{ij} + \eta_i = 0 \quad i = 1, \dots, n$$

- (d) Show that if a set of Marshallian demand functions satisfies homogeneity, symmetry, and Engel aggregation they will also satisfy Cournot aggregation.
5. Show that if complements and substitutes are defined in terms of Marshallian demand derivatives, goods could be, say, complements on the basis of the sign of $\partial D_i / \partial p_j$, and substitutes on the basis of the sign of $\partial D_j / \partial p_i$. Give precise conditions under which this occurs.
6. Show that if the utility function $u(x)$ is an ordinal representation of preferences no restrictions can be placed on the signs of $\partial^2 u^*(p, M) / \partial M^2 = \partial \lambda / \partial M$ and $\partial^2 u^*(p, M) / \partial M \partial p_i = \partial \lambda / \partial p_i$. (Hint: consider positive monotonic transformations of the utility function $G(x) = g(u(x))$, $g' > 0$). Interpret the result. Is it possible to find a numerical representation of preferences $u(x)$ such that the marginal utility of income $\partial u^* / \partial M$ is constant with respect to all prices and income?

C. Measuring the benefits of price changes

We often wish to measure the benefit to consumers of a change in the price of a commodity. The price change may result, for example, from changes in tariffs on imported goods, or in the rate of purchase tax, and we may want to estimate the effect on consumers' welfare for public policy purposes. We know that a change in a price will alter the feasible set confronting a consumer, that a new optimal bundle of goods will result, and that the consumer will be on a new indifference curve. In the case of a price fall the consumer will be better off in the sense that he prefers the new bundle to the initial one. How can we measure this benefit? One suggestion might be by the change in the utility level of the consumer. But the utility functions we have used in the theory of the consumer are ordinal. Hence no significance attaches to the size of utility differences, only to their sign. This means that an ordinal utility measure would be essentially arbitrary. Furthermore, such utility measures would not be comparable among different individuals and we could not add utility differences for a measure of total benefit to all consumers.